Jose Sanchez Gonzalez

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Professional Summary

AI/ML engineer with hands-on experience across computer vision, 3D perception, and NLP — from quantified depth-estimation research for robotic pruning to from-scratch transformers and production-style RAG systems, plus an in-progress humanoid robotics build (ROS2, Isaac Sim, reinforcement learning).

Education

Oregon State University

- **Bachelor of Science in Computer Science** (Applied Route, Data Science), Math Minor — **Graduated - Jun 2025**
- **Masters in Artificial Intelligence** — **Graduated - Jun 2026**

Technical Skills & Relevant Coursework

Graduate-Level Courses: St/Cloud App Development: Docker, Redis, MySQL. Deep Learning: Autoencoders, CNNs, Point-Transformers. Natural Language Processing: LSTMs, Auto-Regressive Transformer, Attentive Bidirectional GRU Translators.
Undergraduate-Level Courses: Database Management: SQL, MongoDB, C. Machine Learning & Data Mining: Python, PyTorch, scikit-learn, Pandas. Mobile Software Dev.: Kotlin, Android Studio, Java.

Projects

Vision-Based Metric Depth Estimation for Robotic Pruning | *Python, PyTorch, DINOv2, CNNs, Blender, OpenCV*

- Built a synthetic-orchard depth estimation pipeline using Blender RGB, depth maps, masks, camera intrinsics and extrinsics for robotic pruning.
- Compared monocular, calibrated, fine-tuned, and multi-view RGB-D depth models for dormant apple tree scenes.
- Improved validation RMSE from 0.0550 m to 0.0445 ± 0.0057 m using DINOv2 RGB+D multi-view refinement, while analyzing deployment latency.
- Profiled deployment latency: the multi-view model required 484.7 ms per 6-view group (80.8 ms amortized per depth map) vs. 333.6 ms per frame for monocular — higher accuracy at the cost of a slower minimum inference unit.

DoM: Decoder-Only Transformer for Math Word Problem Translation | *Python, PyTorch, NLP, Transformers*

- Co-developed a decoder-only seq2seq transformer, built from scratch in PyTorch, that translates math word problems into executable computational graphs on the MathQA dataset (~30k training samples), including custom vocabularies and dataloader.
- Designed a constrained decoding method (smart_sample) guaranteeing valid DAG construction, reaching 77.45% exact-match accuracy, 84.78% BLEU, and 2.72 graph edit distance on the test set.

MetaNaviT: AI-Powered File Organizer | *Python, FastAPI, PostgreSQL, pgvector, LlamaIndex, Ollama, Docker, Linux*

- Developed a file organization and retrieval tool that extracts, indexes, and transforms data from local directories, cloud storage, web content, and databases.
- Used LLMs, pgvector, and retrieval-augmented generation to support semantic search, metadata extraction, hierarchical indexing, and automated file structuring.

Point Cloud Classification | *Python, PyTorch, LiDAR, Deep Learning*

- Designed point-cloud learning models inspired by PointNet and PointConvFormer to classify 3D objects while handling permutation invariance and spatial relationships.
- Trained and evaluated models on ModelNet and ShapeNet datasets using point-based learning and attention mechanisms to improve classification performance.

Experience

Oregon State University **Corvallis, OR**
Graduate Teaching Assistant *Sept 2025 – Present*

- **Teaching:** Supported Operating Systems I (C++) and Computational Methods with COMSOL, covering concurrency, memory, file systems, numerical methods, and simulation workflows.
- **Instruction:** Led 2-hour studio sessions for 25+ students, held office hours, graded assignments, and provided programming, debugging, and modeling support.

Oregon State University **Corvallis, OR**
Math and Computer Science Tutor *Nov 2022 – June 2025*

- Tutored and mentored 50+ students in math, programming, and computer science fundamentals through one-on-one support, adapting explanations to different learning styles to strengthen confidence in technical coursework.

Leadership / Extracurricular

- **Berkeley Humanoid Robot Build Project** (in progress): Independently building the open-source Berkeley Humanoid Lite — hands-on mechanical assembly, soldering, and CAD modeling — and integrating ROS2 and Isaac Sim for motion control and reinforcement learning.
- **ICPC (International Collegiate Programming Contest) Club**: Participated in programming contests and coding practice sessions.
- **Jujitsu Club**: Practiced regularly and contributed to club activities and events.